Proposed Plan – Track C: Advanced Swing-Up

Project Title:
Energy-Based Swing-Up and LQR Stabilization of a Cart-Pole Inverted Pendulum

Our team proposes to design and build a cart pole inverted pendulum system based on the mechanical concept shown in the reference video. The system will consist of a motor-driven cart moving horizontally along a track, with a freely rotating pendulum mounted on the cart.

The main objective is to automatically move the pendulum from its natural hanging position to the upright inverted position, then smoothly catch and stabilize it. The project will focus on achieving a fast and repeatable swing-up while minimizing the Track C judging metrics of 10% to 90% rise time and percentage overshoot.

Control Approach

The system will use two control modes such as:

1. Energy-Based Swing-Up Control
When the pendulum starts in the hanging position, an energy-based controller will move the cart back and forth to gradually increase the pendulum's mechanical energy. The controller will continue adding energy until the pole approaches the upright position.

2. LQR Stabilization Control
When the pendulum enters a predefined region near the upright position, the system will automatically transition to an LQR controller. The LQR controller will use the measured system states to catch the pendulum and maintain stable upright balance.

The overall control sequence will be:

Hanging Position → Energy-Based Swing-Up → Automatic Transition → LQR Catch → Stable Upright Position

Proposed Hardware

The prototype will use a linear cart-pole configuration with components such as:

- Microcontroller such as an STM32 or Arduino Uno
- DC motor with encoder
- Rotary encoder for pendulum angle measurement
- Linear rail and belt
- Lightweight pendulum arm
- Bearings and pivot hardware
- DC power supply
- A crylic, wood, or aluminum structural components
- Ultrasonic sensor

Measurements and Feedback

The controller will monitor four primary system variables:

- Pendulum angle
- Pendulum angular velocity
- Cart position
- Cart velocity

These measurements will be processed by the microcontroller in real time to determine the required motor command.

Development Part

Our development process will include:

1. Build and test the cart-pole mechanical system.
2. Install and calibrate the pendulum and cart encoders.
3. Develop reliable motor position and velocity control.
4. Implement and tune LQR stabilization near the upright position.
5. Implement the energy-based swing-up controller.
6. Develop an automatic transition between swing-up and stabilization.
7. Tune the switching conditions to achieve a smooth catch.
8. Optimize the controller to reduce rise time and percentage overshoot.
9. Perform repeated experimental trials.
10. Record pendulum angle, cart position, motor command, and time for performance evaluation.

Performance Evaluation

The project will be evaluated using the official Track C metrics.

Rise Time:
The time required for the pendulum response to move from 10% to 90% of its final steady-state angle.

Percentage Overshoot:
The amount by which the pendulum moves beyond the desired upright position before settling.

Multiple trials will be performed to evaluate repeatability and identify the controller settings that provide the best combination of fast swing-up and low overshoot.

Final Goal

The final goal is a completely hands-free demonstration in which the system starts with the pendulum hanging downward and automatically performs:

Hanging → Swing-Up → Smooth Catch → Stable Upright Balance